

2020년 1학기 C프로그래밍 2반 (40121-2)

실기시험2 풀이

6/5 (금요일) 10:00~11:50

아래 제시된 프로그램은 여러 가지 가능한 답 중 하나일 뿐이며, 요구되는 기능을 동일하게 수행하는 프로그램을 다른 방법으로 작성할 수 있다.

프로그램 소스 코드

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

#define MAX_WORD_COUNT 1000
#define MAX_LINE_LENGTH 10000
#define MAX_WORD_LENGTH 101
#define OUT_FORMAT "%04d: %4d %3d %s\n"

typedef struct wordEntry
{
    char *word;
    int count;
} Word;

int isAlphabet(char c)
{
    return (((c >= 'A' && c <= 'Z') || (c >= 'a' && c <= 'z')) ? 1 : 0);
}

/* 단어 길이가 2이상이고 단어에 영어 알파벳이 하나라도 포함되어 있으면 유효한 단어임 */
char *checkValidWord(char *p)
{
    int i;

    if (p == (char *)NULL || *p == '\0' || strlen(p) <= 1)
        return NULL;

    for (i = 0; p[i] != '\0'; i++)
        if (isAlphabet(p[i]))
            return p;

    return NULL;
}

int findWord(char *buf, Word *db, int wordCount)
{
    int i;

    if (wordCount == 0)
        return -1;
    for (i = 0; i < wordCount; i++)
        if (strcmp(buf, db[i].word) == 0)
            return i;
    return -1;
}

int saveSingleWord(char *p, Word *db, int wordCount)
{
    static char wordBuffer[MAX_WORD_COUNT][MAX_WORD_LENGTH];
    static int wordBufferCount = 0;
    int index;

    if (p == (char *)NULL || *p == '\0')
        return wordCount;
}
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    if ((index = findWord(p, db, wordCount)) < 0) //the word not found in DB
    {
        db[wordCount].word = wordBuffer[wordBufferCount++];
        strcpy(db[wordCount].word, p);
        db[wordCount].count = 1;
        wordCount++;
    }
    else // the word found in DB
    {
        db[index].count++;
    }

    return wordCount;
}

int getWords(Word *db, int maxNWords)
{
    int wordCount = 0;
    char buffer[MAX_LINE_LENGTH];
    char *seps = " \t\n!\"#$%&'()*+,-./:;<=>?@[\\]^_`{|}~";
    char *word;

    while (fgets(buffer, MAX_LINE_LENGTH, stdin) != NULL)
    {
        if (strlen(buffer) == MAX_LINE_LENGTH - 1)
            fprintf(stderr, "Warning: getWords: a long line with length %d detected!\n",
                strlen(buffer));

        if ((word = strtok(buffer, seps)) != NULL)
        {
            if ((wordCount = saveSingleWord(
                checkValidWord(word), db, wordCount)) == maxNWords)
            {
                fprintf(stderr, "Warning: getWords: max word count reached!\n");
                return(wordCount);
            }
            while ((word = strtok(NULL, seps)) != NULL)
            {
                if ((wordCount = saveSingleWord(
                    checkValidWord(word), db, wordCount)) == maxNWords)
                {
                    fprintf(stderr, "Warning: getWords: max word count reached!\n");
                    return(wordCount);
                }
            }
        }

        return(wordCount);
    }
}

void copyWordEntry(Word *dst, Word *src)
{
    dst->word = src->word;
    dst->count = src->count;
}

void swapWordEntry(Word *a, Word *b)
{
    Word tmp;

    copyWordEntry(&tmp, a);
    copyWordEntry(a, b);
    copyWordEntry(b, &tmp);
}

/* perform bubble sort by words in dictionary order */
void sortWords(Word *db, int n)
{
    int i, j;

    for (i = 0; i < n - 1; i++)
        for (j = 0; j < (n - i) - 1; j++)
            if (strcmp(db[j].word, db[j + 1].word) > 0)

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        swapWordEntry(&db[j], &db[j + 1]);
    }

    /* perform bubble sort by appearance count in decreasing order */
    void sortWordsByCount(Word *db, int n)
    {
        int i, j;

        for (i = 0; i < n - 1; i++)
            for (j = 0; j < (n - i) - 1; j++)
                if (db[j].count < db[j + 1].count)
                    swapWordEntry(&db[j], &db[j + 1]);
    }

    /* perform bubble sort by word length in increasing order */
    void sortWordsByLength(Word *db, int n)
    {
        int i, j;

        for (i = 0; i < n - 1; i++)
            for (j = 0; j < (n - i) - 1; j++)
                if (strlen(db[j].word) > strlen(db[j + 1].word))
                    swapWordEntry(&db[j], &db[j + 1]);
    }

    void putWords(Word *db, int wordCount)
    {
        int i;

        printf("Number of different words = %d\n", wordCount);
        for (i = 0; i < wordCount; i++)
            printf(OUT_FORMAT, i + 1, db[i].count, strlen(db[i].word), db[i].word);
    }

    int main(void)
    {
        Word allWords[MAX_WORD_COUNT] = { 0 };
        int wordCount, groupStartIndex, groupCount;
        int idx, maxIdx, cnt;

        wordCount = getWords(allWords, MAX_WORD_COUNT);

        // Sort the words by their appearance counts in decreasing order
        sortWordsByCount(allWords, wordCount);

        // For each group of words with equal appearance counts
        for (groupStartIndex = 0; groupStartIndex < wordCount; groupStartIndex += groupCount)
        {
            // Find the boundary of the next group of words with equal appearance counts
            for (groupCount = 1; groupStartIndex + groupCount < wordCount; groupCount++)
                if (allWords[groupStartIndex + groupCount].count !=
                    allWords[groupStartIndex].count)
                    break;

            if (groupCount > 1)
            {
                // Within the word group with equal appearance counts,
                // sort words by their lengths in increasing order
                sortWordsByLength(&allWords[groupStartIndex], groupCount);

                // For each subgroup of words with equal appearance counts and equal lengths
                for (idx = groupStartIndex, maxIdx = groupStartIndex + groupCount;
                    idx < maxIdx;
                    idx += cnt)
                {
                    // Within the word group with equal appearance counts,
                    // find the boundary of next subgroup of words with equal lengths
                    for (cnt = 1; idx + cnt < maxIdx; cnt++)
                        if (strlen(allWords[idx + cnt].word) != strlen(allWords[idx].word))
                            break;

                    // Sort the words

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        // in each subgroup of words with equal appearance counts & equal lengths
        // in dictionary order
        if (cnt > 1)
            sortWords(&allWords[idx], cnt);
    }
}

putWords(allWords, wordCount);

fprintf(stderr, "Processing completed!!\n"); return 0;
}

```

시험용 아스키 텍스트 파일 trump.txt: https://en.wikipedia.org/wiki/Donald_Trump의 내용을 메모장으로 copy & paste하여 만든 파일임. 한 줄이 매우 길기 때문에 개행문자들을 적절히 넣고, "-----"로만 구성된 줄을 문단 마지막마다 넣어 내용을 조정하였음.

trump - 메모장

파일(F) 편집(E) 서식(O) 보기(V) 도움말(H)

Donald John Trump (born June 14, 1946) is the 45th and current president of the United States. Before entering politics, he was a businessman and television personality.

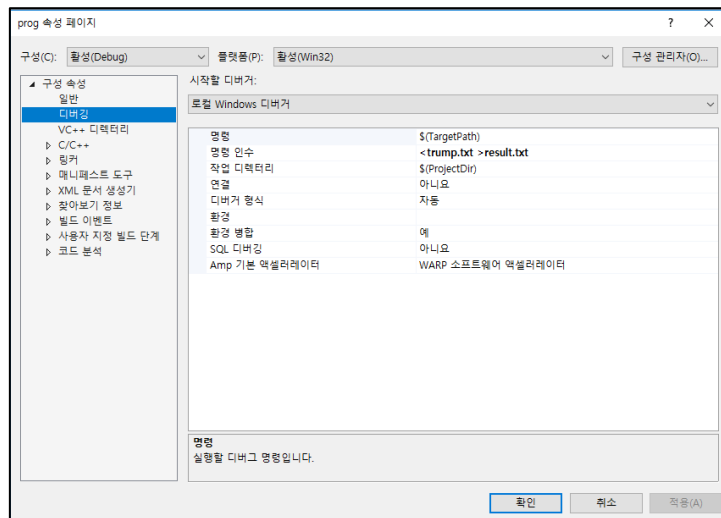
Trump was born and raised in Queens, a borough of New York City, and received a bachelor's degree in economics from the Wharton School. He took charge of his family's real-estate business in 1971, renamed it The Trump Organization, and expanded its operations from Queens and Brooklyn into Manhattan. The company built or renovated skyscrapers, hotels, casinos, and golf courses. Trump later started various side ventures, mostly by licensing his name. He produced and hosted The Apprentice, a reality television series, from 2003 to 2015. As of 2020, Forbes estimated his net worth to be \$2.1 billion.[a]

Trump entered the 2016 presidential race as a Republican and defeated 16 other candidates in the primaries. His political positions have been described as populist, protectionist, and nationalist. Despite not being favored in most forecasts, he was elected over Democratic nominee Hillary Clinton, while still losing the popular vote. He became the oldest first-term U.S. president,[b] and the first without prior military or government service. His election and policies have sparked numerous protests. Trump has made many false or misleading statements during his campaign and presidency. The statements have been documented by fact-checkers, and the media have widely described the phenomenon as unprecedented in American politics. Many of his comments and actions have been characterized as racially charged or racist.

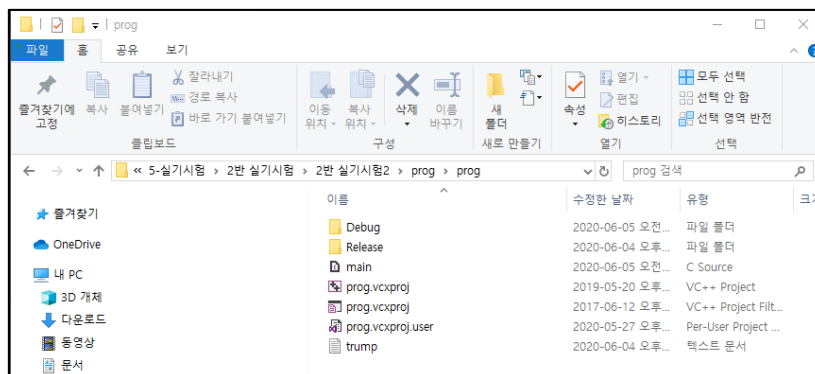
During his presidency, Trump ordered a travel ban on citizens from several Muslim-majority countries, citing security concerns; after legal challenges, the Supreme Court upheld the policy's third revision. He enacted a tax-cut package for individuals and businesses, rescinding the individual health insurance mandate penalty. He appointed Neil Gorsuch and Brett Kavanaugh to the Supreme Court. In foreign policy, Trump has pursued an America First agenda, withdrawing the U.S. from the Trans-Pacific Partnership trade negotiations, the Paris Agreement on climate change, and the Iran nuclear deal. During increased tensions with Iran, he ordered the killing of Iranian general Qasem Soleimani. He imposed import tariffs triggering a trade war with China, recognized Jerusalem as the capital of Israel, and withdrew U.S. troops from northern Syria to avoid Turkey's offensive on American-allied Kurds. Trump met three times with North Korea's leader Kim Jong-un, but talks on denuclearization broke down in 2019.

A special counsel investigation led by Robert Mueller found that Trump and his campaign welcomed and encouraged Russian foreign interference in the 2016 presidential election under the belief that it would be politically advantageous, but did not find sufficient evidence to press charges of criminal conspiracy or coordination with Russia. Mueller also investigated Trump for obstruction of justice, and his report neither indicted nor exonerated Trump on that count. A 2019 House of Representatives impeachment inquiry found that Trump solicited foreign interference in the 2020 U.S. presidential election from Ukraine to help his re-election bid and then obstructed the inquiry itself. The House impeached Trump on December 18, 2019, for abuse of power and obstruction of Congress. The Senate acquitted him of both charges on February 5, 2020.

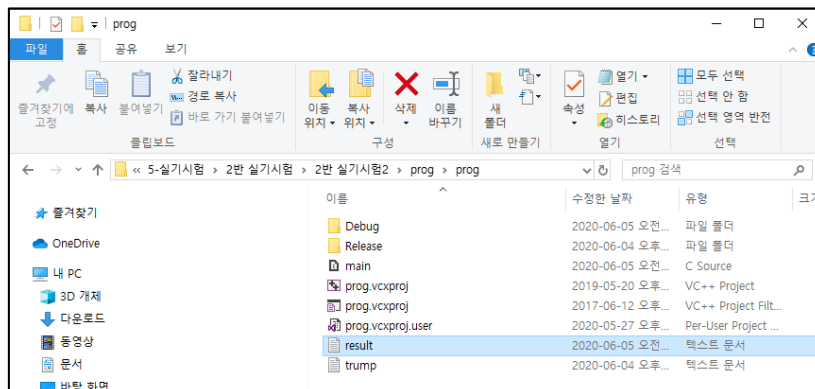
VS2017의 프로젝트 속성 설정: 입출력 redirection을 위한 프로젝트 속성 설정. stdin 대신 trump.txt로부터 입력 받고 stdout 대신 result.txt로 출력하도록 함



실행 전 입력파일 trump.txt의 위치: 프로젝트 폴더 내



실행 후 출력파일 result.txt의 위치: 프로젝트 폴더 내



실행 결과 출력파일 result.txt를 메모장으로 확인:

result - 메모장	result - 메모장	result - 메모장
파일(F) 편집(E) 서식(O) 보기(V) 도움말(H)	파일(F) 편집(E) 서식(O) 보기(V)	파일(F) 편집(E) 서식(O) 보기(V)
Number of different words = 312	0061: 1 3 bid	0122: 1 5 third
0001: 24 3 and	0062: 1 3 did	0123: 1 5 three
0002: 24 3 the	0063: 1 3 him	0124: 1 5 times
0003: 14 5 Trump	0064: 1 3 its	0125: 1 5 under
0004: 13 2 of	0065: 1 3 led	0126: 1 5 while
0005: 9 2 in	0066: 1 3 met	0127: 1 5 worth
0006: 9 3 his	0067: 1 3 net	0128: 1 5 would
0007: 7 2 on	0068: 1 3 nor	0129: 1 6 Before
0008: 7 4 from	0069: 1 3 war	0130: 1 6 Donald
0009: 6 2 He	0070: 1 4 45th	0131: 1 6 Forbes
0010: 6 2 to	0071: 1 4 City	0132: 1 6 Israel
0011: 6 3 The	0072: 1 4 John	0133: 1 6 Robert
0012: 5 2 as	0073: 1 4 June	0134: 1 6 Russia
0013: 5 2 or	0074: 1 4 Many	0135: 1 6 School
0014: 5 4 have	0075: 1 4 Neil	0136: 1 6 Senate
0015: 4 4 that	0076: 1 4 York	0137: 1 6 States
0016: 4 4 with	0077: 1 4 also	0138: 1 6 Turkey
0017: 3 2 by	0078: 1 4 both	0139: 1 6 United
0018: 3 2 he	0079: 1 4 deal	0140: 1 6 agenda
0019: 3 3 for	0080: 1 4 down	0141: 1 6 became
0020: 3 3 was	0081: 1 4 find	0142: 1 6 belief
0021: 3 4 been	0082: 1 4 golf	0143: 1 6 change
0022: 3 7 foreign	0083: 1 4 help	0144: 1 6 charge
0023: 3 8 election	0084: 1 4 into	0145: 1 6 citing
0024: 3 12 presidential	0085: 1 4 made	0146: 1 6 degree
0025: 2 2 be	0086: 1 4 many	0147: 1 6 during
0026: 2 2 it	0087: 1 4 most	0148: 1 6 family
0027: 2 3 His	0088: 1 4 name	0149: 1 6 health
0028: 2 3 but	0089: 1 4 over	0150: 1 6 hosted
0029: 2 3 has	0090: 1 4 race	0151: 1 6 hotels
0030: 2 3 not	0091: 1 4 side	0152: 1 6 import
0031: 2 4 Iran	0092: 1 4 then	0153: 1 6 itself
0032: 2 4 born	0093: 1 4 took	0154: 1 6 leader
0033: 2 5 Court	0094: 1 4 vote	0155: 1 6 losing
0034: 2 5 House	0095: 1 5 Brett	0156: 1 6 mostly
0035: 2 5 found	0096: 1 5 China	0157: 1 6 oldest
0036: 2 5 trade	0097: 1 5 First	0158: 1 6 racist
0037: 2 6 During	0098: 1 5 Korea	0159: 1 6 raised
0038: 2 6 Queens	0099: 1 5 Kurds	0160: 1 6 report
0039: 2 6 policy	0100: 1 5 North	0161: 1 6 series
0040: 2 7 Mueller	0101: 1 5 Paris	0162: 1 6 travel
0041: 2 7 Supreme	0102: 1 5 Qasem	0163: 1 6 troops
0042: 2 7 charges	0103: 1 5 Syria	0164: 1 6 upheld
0043: 2 7 inquiry	0104: 1 5 abuse	0165: 1 6 widely
0044: 2 7 ordered	0105: 1 5 after	0166: 1 7 America
0045: 2 8 campaign	0106: 1 5 avoid	0167: 1 7 Clinton
0046: 2 8 politics	0107: 1 5 being	0168: 1 7 Despite
0047: 2 9 described	0108: 1 5 broke	0169: 1 7 Gorsuch
0048: 2 9 president	0109: 1 5 built	0170: 1 7 Hillary
0049: 2 10 presidency	0110: 1 5 count	0171: 1 7 Iranian
0050: 2 10 statements	0111: 1 5 false	0172: 1 7 Jong-un
0051: 2 10 television	0112: 1 5 first	0173: 1 7 Russian
0052: 2 11 obstruction	0113: 1 5 later	0174: 1 7 Ukraine
0053: 2 12 interference	0114: 1 5 legal	0175: 1 7 Wharton
0054: 1 2 As	0115: 1 5 media	0176: 1 7 actions
0055: 1 2 In	0116: 1 5 other	0177: 1 7 billion
0056: 1 2 an	0117: 1 5 power	0178: 1 7 borough
0057: 1 2 is	0118: 1 5 press	0179: 1 7 capital
0058: 1 3 Kim	0119: 1 5 prior	0180: 1 7 casinos
0059: 1 3 New	0120: 1 5 still	0181: 1 7 charged
0060: 1 3 ban	0121: 1 5 talks	0182: 1 7 climate

result - 메모장					result - 메모장				
파일(F)	편집(E)	서식(O)	보기(V)		파일(F)	편집(E)	서식(O)	보기(V)	도움말(H)
0183:	1	7	company		0244:	1	8	withdrew	
0184:	1	7	counsel		0245:	1	9	Agreement	
0185:	1	7	courses		0246:	1	9	Jerusalem	
0186:	1	7	current		0247:	1	9	Kavanaugh	
0187:	1	7	elected		0248:	1	9	Manhattan	
0188:	1	7	enacted		0249:	1	9	Soleimani	
0189:	1	7	entered		0250:	1	9	acquitted	
0190:	1	7	avored		0251:	1	9	appointed	
0191:	1	7	general		0252:	1	9	countries	
0192:	1	7	imposed		0253:	1	9	economics	
0193:	1	7	justice		0254:	1	9	estimated	
0194:	1	7	killing		0255:	1	9	forecasts	
0195:	1	7	mandate		0256:	1	9	impeached	
0196:	1	7	neither		0257:	1	9	increased	
0197:	1	7	nominee		0258:	1	9	insurance	
0198:	1	7	nuclear		0259:	1	9	licensing	
0199:	1	7	package		0260:	1	9	offensive	
0200:	1	7	penalty		0261:	1	9	political	
0201:	1	7	popular		0262:	1	9	positions	
0202:	1	7	pursued		0263:	1	9	primaries	
0203:	1	7	reality		0264:	1	9	renovated	
0204:	1	7	renamed		0265:	1	9	solicited	
0205:	1	7	service		0266:	1	10	Apprentice	
0206:	1	7	several		0267:	1	10	Democratic	
0207:	1	7	sparked		0268:	1	10	Republican	
0208:	1	7	special		0269:	1	10	businesses	
0209:	1	7	started		0270:	1	10	candidates	
0210:	1	7	tariffs		0271:	1	10	challenges	
0211:	1	7	tax-cut		0272:	1	10	conspiracy	
0212:	1	7	various		0273:	1	10	documented	
0213:	1	7	without		0274:	1	10	encouraged	
0214:	1	8	American		0275:	1	10	exonerated	
0215:	1	8	Brooklyn		0276:	1	10	first-term	
0216:	1	8	Congress		0277:	1	10	government	
0217:	1	8	December		0278:	1	10	individual	
0218:	1	8	February		0279:	1	10	misleading	
0219:	1	8	bachelor		0280:	1	10	obstructed	
0220:	1	8	business		0281:	1	10	operations	
0221:	1	8	citizens		0282:	1	10	phenomenon	
0222:	1	8	comments		0283:	1	10	recognized	
0223:	1	8	concerns		0284:	1	10	rescinding	
0224:	1	8	criminal		0285:	1	10	sufficient	
0225:	1	8	defeated		0286:	1	10	triggering	
0226:	1	8	entering		0287:	1	11	Partnership	
0227:	1	8	evidence		0288:	1	11	businessman	
0228:	1	8	expanded		0289:	1	11	impeachment	
0229:	1	8	indicted		0290:	1	11	individuals	
0230:	1	8	military		0291:	1	11	nationalist	
0231:	1	8	northern		0292:	1	11	personality	
0232:	1	8	numerous		0293:	1	11	politically	
0233:	1	8	policies		0294:	1	11	re-election	
0234:	1	8	populist		0295:	1	11	real-estate	
0235:	1	8	produced		0296:	1	11	skyscrapers	
0236:	1	8	protests		0297:	1	11	withdrawing	
0237:	1	8	racially		0298:	1	12	Organization	
0238:	1	8	received		0299:	1	12	advantageous	
0239:	1	8	revision		0300:	1	12	coordination	
0240:	1	8	security		0301:	1	12	investigated	
0241:	1	8	tensions		0302:	1	12	negotiations	
0242:	1	8	ventures		0303:	1	13	Trans-Pacific	
0243:	1	8	welcomed		0304:	1	13	characterized	

끝.